



SGGS Institute of Engg. & Tech., Nanded
Dept. of Electronics & Telecom. Engg.
B. Tech. (EXTC) ISE-1 Examination
Artificial Intelligence and Neural Networks (OEC-EC407)

Note: I. Solve ALL Questions

II. Assume suitable data if necessary

Date: 21 August 2025, Time : 11:00-11:45

Max marks: 20

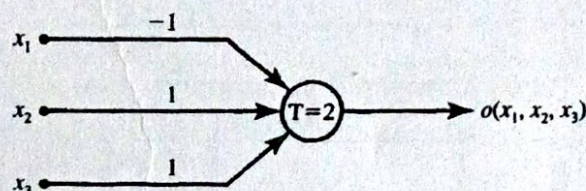
Q.1 a) Discuss and drive the weight updating formula for Delta learning rule. (05)

b) Find the weight vector of a neuron with $\tanh()$ activation function after ONE iterations of training using Delta Learning Rule. The training data is:

$$X_1 = [-1, -3, -2, -1]^T, d_1 = '+1', X_2 = [1, -3, 2, 1]^T, d_2 = '-1', X_3 = [-1, -1, 2, 1]^T, d_3 = '-1'$$

Assume learning constant is $c = 0.5$ and initial weight vector = $[-1, -2, 1, 1]^T$ (05)

Q.2 a) Explain McCulloch and Pitt's model of artificial Neuron (MPN). Find the logic implemented by following circuit of MPNs: (05)



b) Perform Hebbian learning on a single-neuron. The task is to start with an initial weight vector (05)

$$w^1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and use a learning constant $c = 1$ to update the weights using the following four input vectors:

$$x_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, x_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

You need to apply Hebbian learning steps sequentially to find the final weight vector after processing all four input vectors, using a bipolar binary activation function.

Course Outcomes (COs):

1. Understanding the basics of Artificial Intelligence, artificial neural network, their limitations, basic pattern analysis tasks such as classification and clustering, learning and adaptation using the learning rules, implementation of learning rule.
2. Describe the concepts of feed forward neural networks using single layer and multilayer networks to solve classification problem, and its implementation, single layer feedback networks to study the concept of memory using neural networks.
3. Analyze and implement the applications of neural networks in character recognition and control systems.
4. Understand Auto associative neural networks, Pattern storage and retrieval, Hopfield model, recurrent neural networks
5. Analyze Bayesian neural networks, Radial basis function networks
6. Understand self-organizing maps and recent trends in neural networks

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyse, BT5- Evaluate, BT6- Create

Question No	CO No	BT No
1 (a,b)	1	1, 2
2 (a,b)	2	1, 6



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication
In-Semester Examination I
Academic Year: 2025-26 (Semester-I)

Class: Final Year B.Tech.	Course: Data Communication and Networking	Course Code: PEC-EC-403
Date: 19.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	All Questions are compulsory	Marks	CO	BT
Q.1	Explain Open Systems Interconnection (OSI) reference model?	05	CO1	BT2
Q.2	Write short notes on i) Routers ii) Switches.	05	CO2	BT1
Q.3	Explain Co-axial cable in detail?	05	CO2	BT1
Q.4	Compare circuit switching and packet switching?	05	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Describe the components and infrastructure that form the basis for most computer networks
CO 2	Describe the technical aspects of data communications on the Internet
CO 3	Achieve practical experience in network programming.
CO 4	Propose network designs based on case studies in colleges or other institutions
CO 5	Recognize security issues in a network, Design and Evaluate short path algorithm.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE 1 Examination (Semester-I) 2025-26

Electronics and Telecommunication Engineering

**Final Year. B. Tech.
Microwave Engineering
Time:11.00 -11.45**

**PCC-EC-401
Max. Marks :20**

Date:18/08/2025

- Note: i. Attempt any two questions
ii. Assume suitable data if necessary
iii. Use of non-programmable calculator is permitted.

Q.N.		Question	Marks
Q.1	a	Explain reflection coefficient and VSWR of transmission line. What is the relation between them?	5
	b	Derive an expression for characteristics impedance of lossy and lossless transmission line.	5
Q.2	a	Determine the stub length and the distance of the stub from the load. Given that a complex load $Z_L = 50 - j 100$ is to be matched to a 75Ω transmission line using a short- circuited stub. Use smith chart.	5
	b	Following are the parameters of a transmission line: $R = 5 \Omega$, $L = 1 \text{ m H}$, $G = 0.1 \text{ S}$, and $C = 1 \text{ n F}$ at 1 MHz Determine Characteristics impedance, attenuation constant, phase constant and propagation constant.	5
Q.3	a	Derive an expression for input impedance of short circuited and open circuited transmission line.	5
	b	A lossless transmission line of 50Ω is having $V_{\max} = 2.5 \text{ V}$, $V_{\min} = 1 \text{ V}$ when terminated by unknown load find VSWR, γ and Z_L	5



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.

Summer Term Examination 2024-25

Final Year. B. Tech.
Microwave Engineering

Electronics and Telecommunication Engineering
PCC-EC-401

Date: 17/06/2025

Time: 14.30 – 17.30

Max. Marks : 100

- Note: i. Attempt any ten questions.
ii. Assume suitable data if necessary.
iii. Use of non-programmable calculator is permitted.

Q.N.		Question	Marks
Q.1	a	Explain the TE_{10} of rectangular waveguide.	5
	b	Explain the internal structure of IMPATT diode in detail.	5
Q.2	a	Explain internal structure of Gunn diode in detail.	5
	b	Derive an expression for voltage and current equation of transmission lines.	5
Q3	a	Derive an expression for reflection coefficient of transmission line.	5
	b	Explain the structure and working of Helix Travelling Wave Tube in detail.	5
Q4	a	Explain with neat diagram reflex klystron oscillator.	5
	b	Derive an expression for Electric wave equation of microwave waveguide.	5
Q5	a	State and Explain the Poynting theorem in electromagnetic plane waves.	5
	b	Explain various microwave devices in short.	5
Q6	a	Explain microwave cavities.	5
	b	A lossless transmission line of 50Ω is having $V_{\max} = 2.5 \text{ V}$, $V_{\min} = 1 \text{ V}$ when terminated by unknown load find VSWR, γ and Z_L	5
Q7	a	Explain the structure and working of two cavity klystron amplifier.	5
	b	Derive an expression for standing wave ratio of microwave transmission lines.	5
Q8	a	Explain plane wave propagation in lossy media.	5
	b	For transmission lines R, L, G and C components are given by 1Ω , 1 mH , 0.1 nS and $1 \mu\text{F}$ respectively. Find Z_0 , γ , α and β at 1 MHz .	5
Q9	a	Explain internal structure of BARITT diode	5
	b	An air filled rectangular waveguide has inner dimension of $2 \times 3 \text{ cm}^2$. Calculate the wave impedance of TE_{20} mode of propagation in the waveguide at a frequency of 30 GHz . (free space impedance is given by 377Ω)	5
Q10	a	Explain General applications of microwaves in detail.	5
	b	Explain smith chart application in solving transmission line problems.	5
Q11	a	Explain plane wave propagation in free space.	5
	b	Explain microwave hybrid circuits.	5
